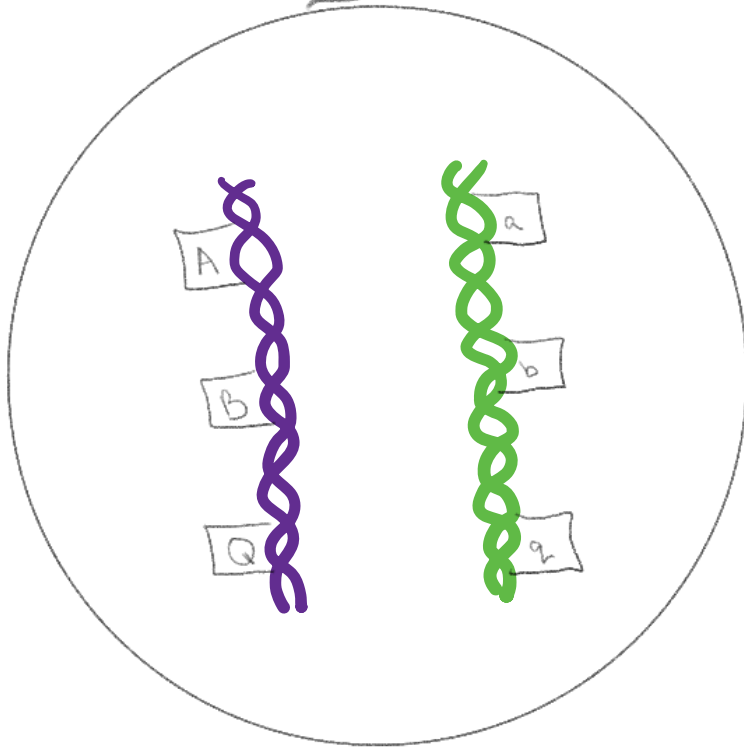


Starting Cell



Alleles

A = sparkly skin
a = scaly skin

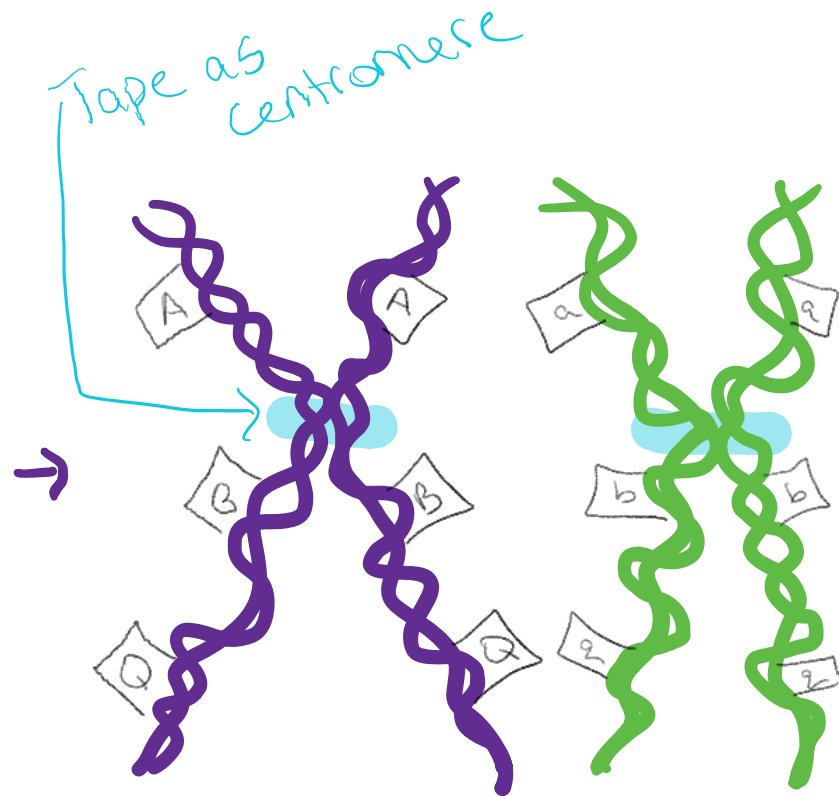
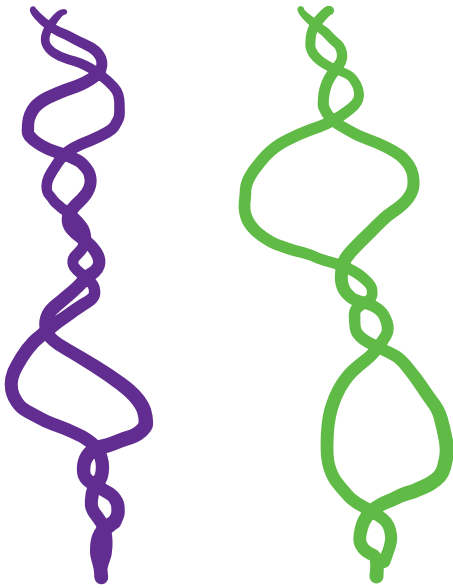
B = long tail
b = short tail

Q = Four eyes
q = one eye

phenotype:

Sparkly skin
Long tail
One eye

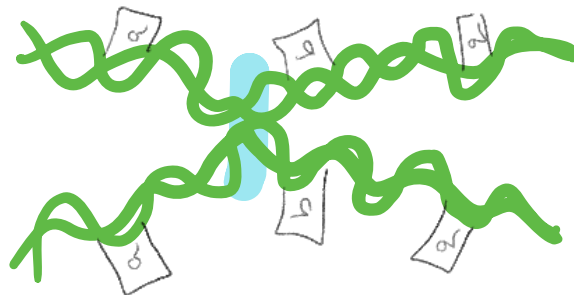
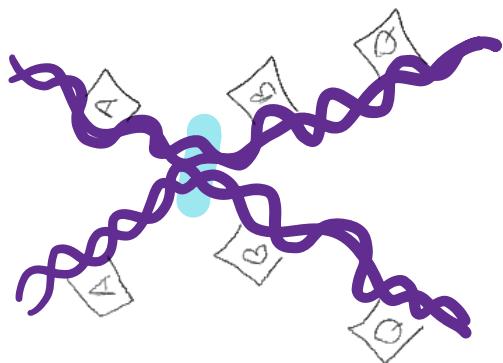
origins of replication



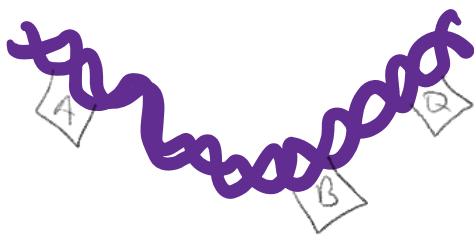
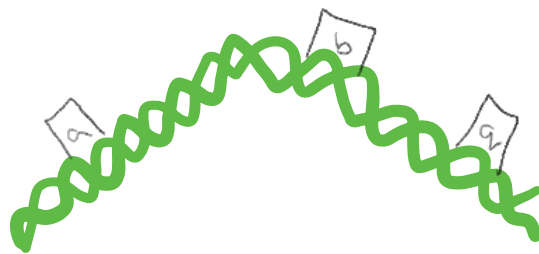
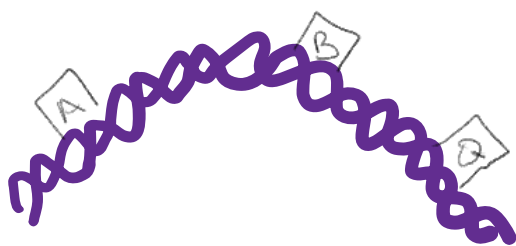
After S-phase

Mitosis

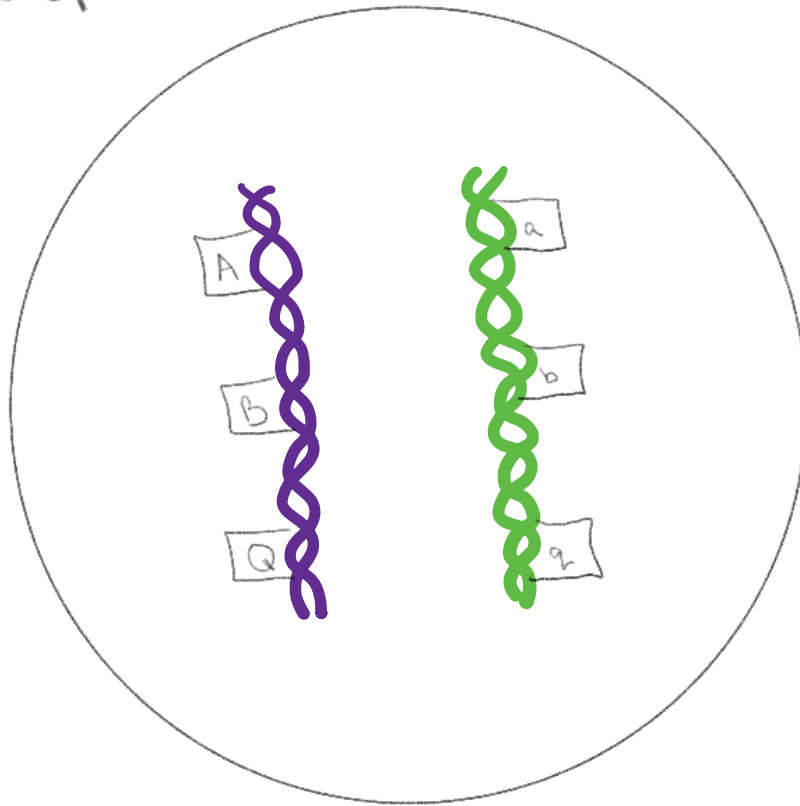
Metaphase



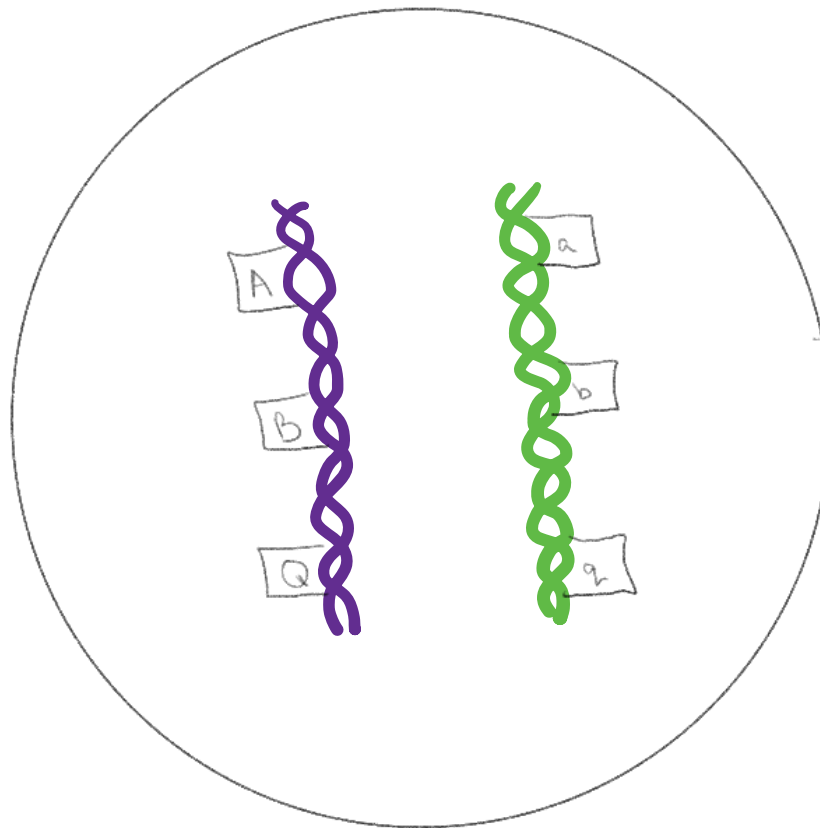
Anaphase



Telophase Complete



phenotype:
Sparkly skin
Long tail
One eye

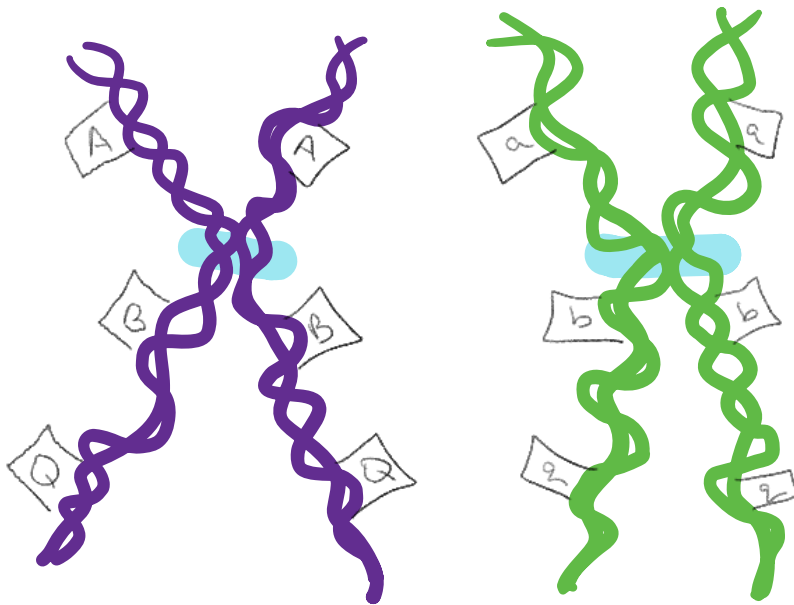


phenotype:
Sparkly skin
Long tail
One eye

How many cells are produced? **2**
What is their ploidy? **Diploid**

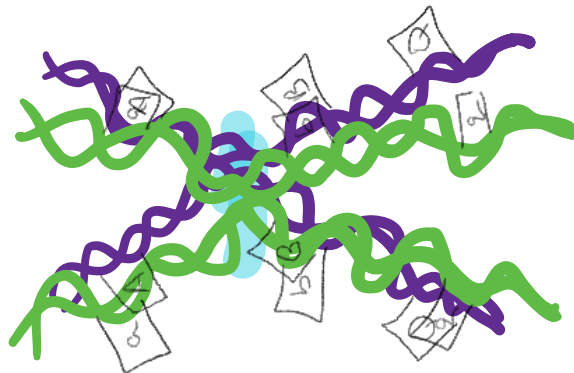
How do they compare to the starting cell? **Identical**

Starting cell after S-phase

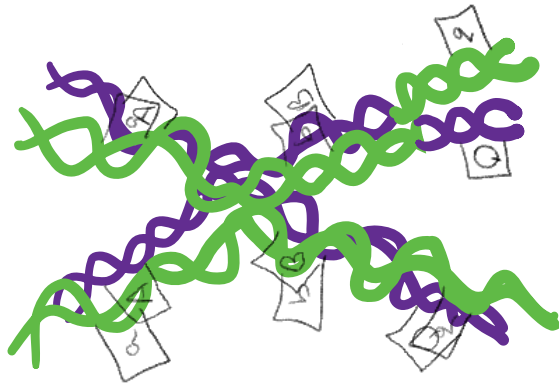


Meiosis

prophase I

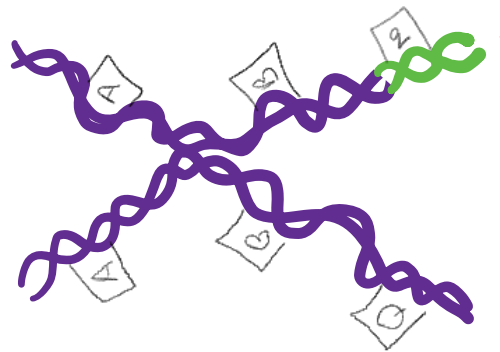


Homology
Search
↓
Synapsis

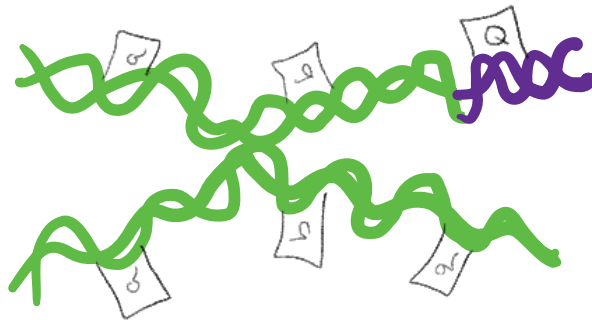


Crossing Over
+ recombination

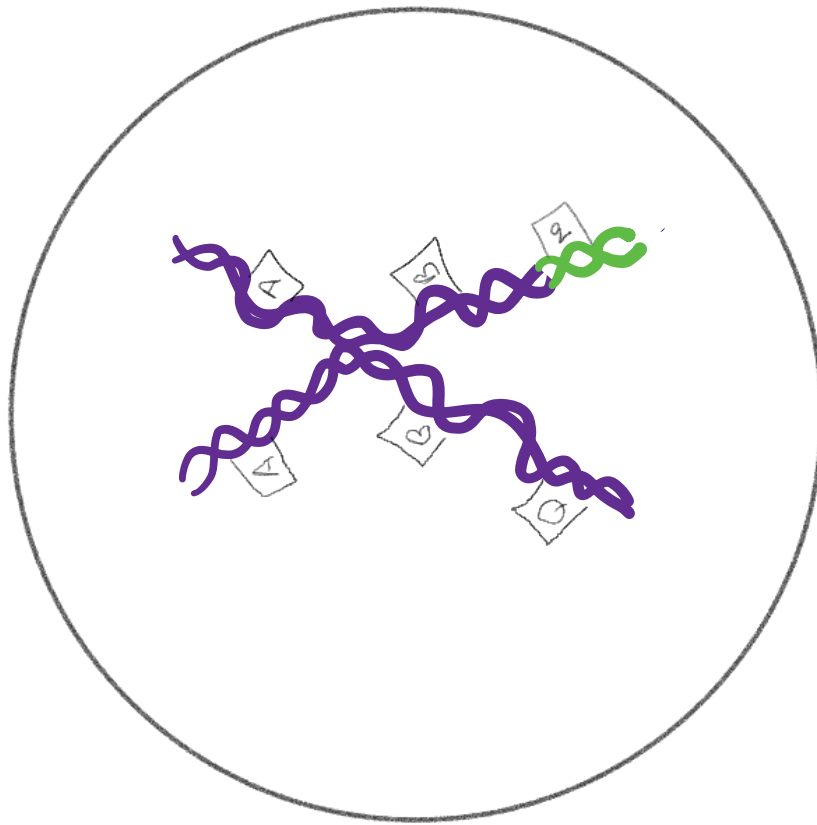
Anaphase I



Homologs
pulled
apart

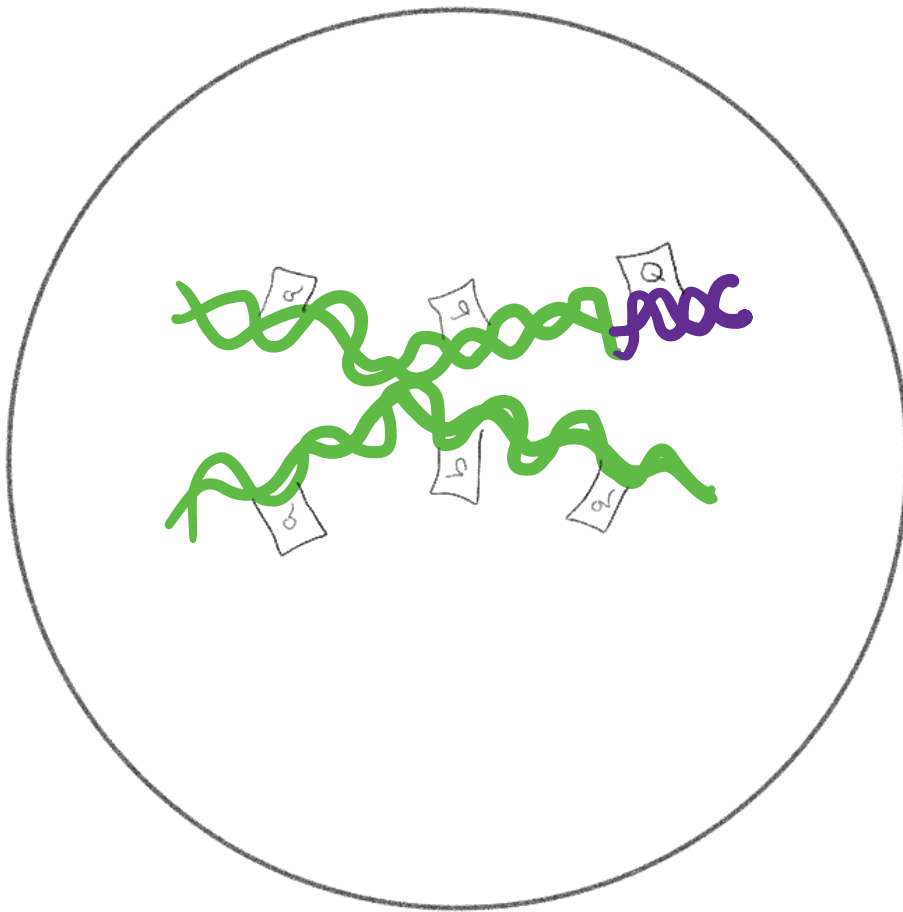


End of Telophase I

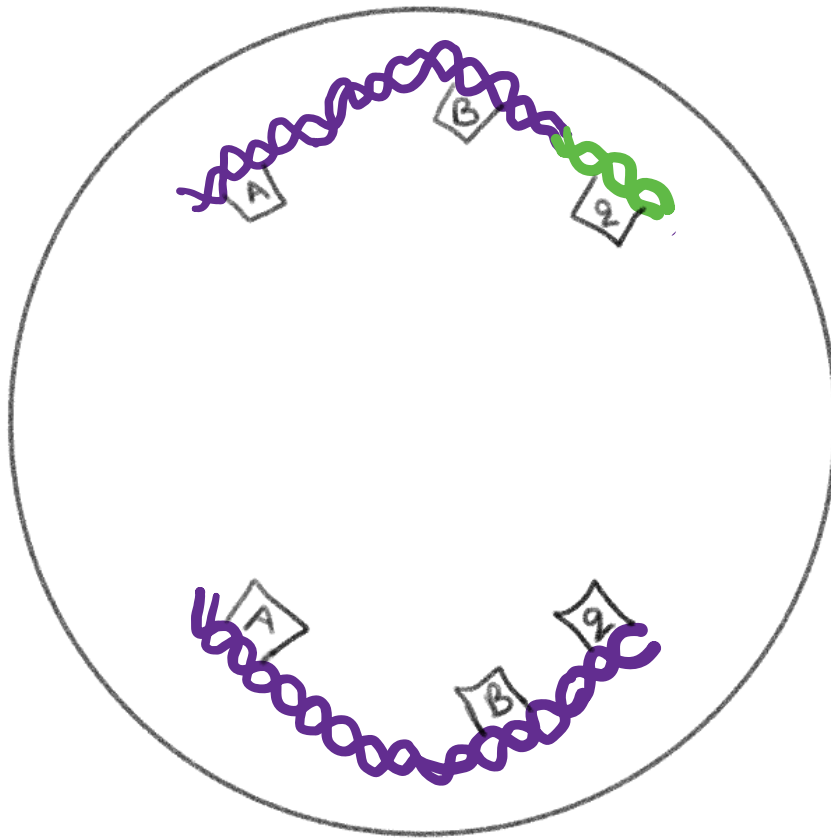


Cells are
now
haploid

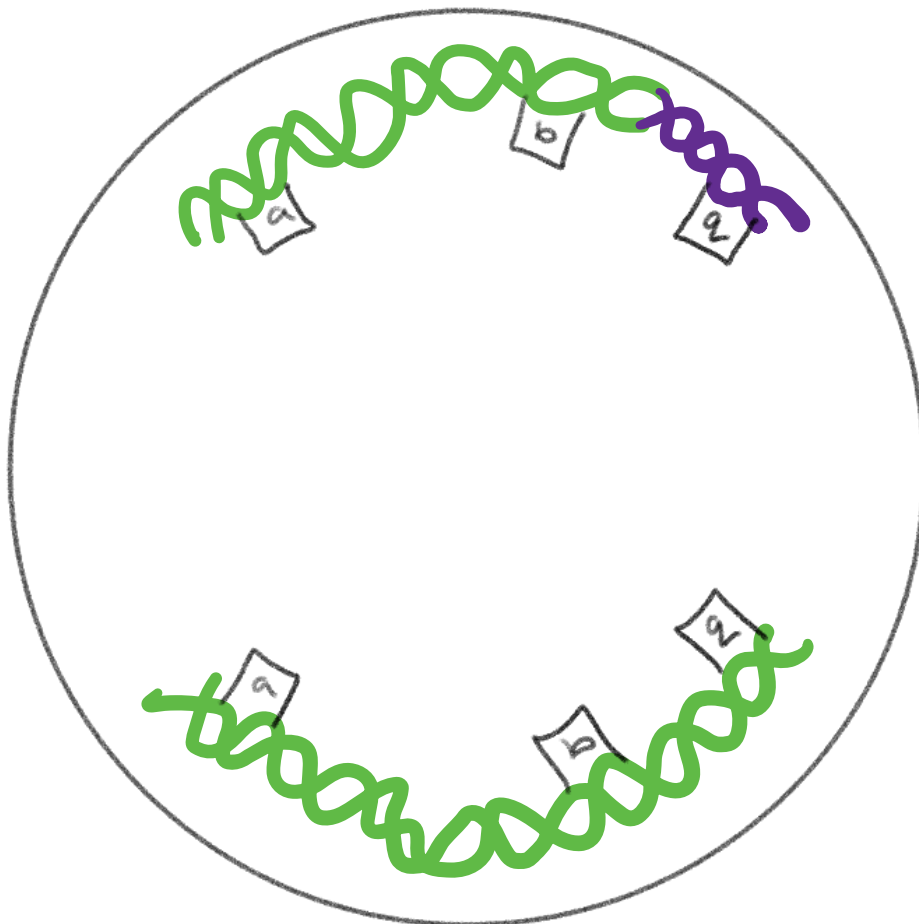
this is
the
reductional
division



Anaphase II



Sister
chromatids
Separate



End of Telophase II

